# Secular Equation and Exact Loss Convolution for Minibatch Heavy-Ball SGD on Quadratics

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**Abstract**

For heavy-ball SGD on a quadratic in $\mathbb{R}^P$,

$$w_{t+1} = \big((1+\beta)I - \eta H_{B_t}\big)w_t - \beta w_{t-1} - \eta\xi_t,$$

with i.i.d. minibatch Hessians $H_{B_t}$ of mean $H \succ 0$, we study the mean-square stability threshold $\eta^\star(\beta)$ and the expected loss at finite $P$. When the minibatch Hessians are formed from random features with independent coordinates in the eigenbasis of $H$ and excess kurtosis $\kappa_{\text{feat}} \geq -2$, $\eta^\star(\beta)$ is the unique root of a rank-one secular equation (Theorem 5), and twice the expected loss, $\mathbb{E}[\langle w_t, Hw_t\rangle]$, obeys an exact discrete convolution equation (Proposition 6). Its kernel sum equals the secular left-hand side (for $\kappa_{\text{feat}} \geq -1$ on the whole stable range), and at $\alpha_{\text{diag}} = 1$ with the large-$n$ pool factor of sampling without replacement it has the closed form of the Volterra kernel norm of Lee et al. [2022]. The reduction uses the threshold for Hessians that need not commute, $\eta^\star(\beta) = 1/\lambda_{\max}(\mathcal{L}_0^{-1/2}\mathcal{B}_\beta\mathcal{L}_0^{-1/2})$ (Theorem 3), obtained independently of Cai et al. [2026, Theorem F.1], which gives the same criterion.

## 1 Introduction

Consider heavy-ball stochastic gradient descent on a quadratic loss in $\mathbb{R}^P$ with mean Hessian $H = \sum_{k=1}^P \lambda_k u_k u_k^\top \succ 0$ ($\lambda_1 \geq \cdots \geq \lambda_P > 0$):

$$w_{t+1} = \big((1+\beta)I - \eta H_{B_t}\big)w_t - \beta w_{t-1} - \eta\xi_t, \qquad \beta \in [0,1),\ \eta > 0, \tag{1}$$

where $(H_{B_t}, \xi_t)$ are i.i.d. across steps with $\mathbb{E}[H_{B_t}] = H$, $\mathbb{E}\|H_{B_t}\|_F^2 < \infty$, $\mathbb{E}[\xi_t] = 0$, $\mathbb{E}[\xi_t\xi_t^\top] = \Sigma_\xi \succeq 0$, and curvature-noise operator

$$\mathcal{V}(X) \triangleq \mathbb{E}\big[(H_{B_t} - H)X(H_{B_t} - H)\big] \quad (X \in \mathbb{S}^P).$$

For SGD without momentum ($\beta = 0$), Mulayoff and Michaeli [2024] characterized the exact mean-square stability threshold at interpolating minima as the reciprocal of the top eigenvalue of a symmetrizable operator on $\mathbb{S}^P$. With momentum $\beta \in (0,1)$, the state space doubles to $(w_t, w_{t-1}) \in \mathbb{R}^{2P}$. In high-dimensional limits where the effective rank $\operatorname{tr}(H)/\lambda_1$ is large, Paquette and Paquette [2021], Lee et al. [2022], and Cheng et al. [2024] showed that the quadratic risk concentrates on a deterministic Volterra integral equation; in particular, the proof of Proposition 2 of Lee et al. [2022] gives a closed-form expression for the norm of the Volterra kernel under orthogonally invariant data. With non-degenerate additive noise, bounded second moments of a linear recursion with i.i.d. coefficients are equivalent to $\rho(\mathbb{E}[F_t \otimes F_t]) < 1$ [Costa et al., 2005, Chapter 3]. For momentum LMS with Gaussian inputs at batch size 1, Roy and Shynk [1990] derive a necessary condition of the form of the secular equation of Section 3, and Velikanov et al. [2023] obtain a loss-convergence condition of that form under their spectrally expressible approximation. With momentum at finite $P$, Andreyev et al. [2026] give the threshold exactly in one dimension and, in $P$

dimensions, expand the Schur complement of the second-moment operator on $2P \times 2P$ matrices for small $\eta$ in the noise-dominated regime, where stability depends only on the effective step $\eta/(1-\beta)$ [the rescaling of Yuan et al., 2016]. They list the precise instability criterion of SGD with momentum as open, also for nonlinear networks; for the linear recursion (1) the exact threshold is not a function of $\eta/(1-\beta)$ alone (Corollary 4). Sharma et al. [2026] give an exact mean-square criterion for outer Nesterov momentum. Concurrently and independently, Cai et al. [2026, Theorem F.1] give the criterion of Theorem 3; their eq. (67) gives Corollary 4 without strictness.

In this note we derive, at finite $P$, a secular equation for the threshold and an exact convolution equation for the loss, and connect them to the high-dimensional Volterra kernel:

1. **Threshold for non-commuting Hessians (Section 2).** Eliminating the lag-one covariance closes the stationary equation on $\Sigma_0$ with a double commutator $[H,[H,\Sigma_0]]$ (Theorem 1); a cone-preserving factorization reduces stability to a symmetric eigenvalue problem on $\mathbb{S}^P$ (Theorem 3), and $\eta^\star(\beta)/(1-\beta)$ increases strictly with $\beta$ (Corollary 4).

2. **Secular equation and finite-$P$ convolution equation for the loss (Section 3).** For random-feature minibatches with independent coordinates in the eigenbasis of $H$ and fourth moment $3+\kappa_{\text{feat}} \geq 1$, the diagonal of $\Sigma_0$ decouples from the off-diagonal and the threshold is the unique root of a scalar secular equation (Theorem 5). Twice the expected loss, $\ell_t = \operatorname{tr}(H\mathbb{E}[w_t w_t^\top])$, satisfies an exact discrete convolution equation at finite $P$ (Proposition 6); under the condition of Proposition 6, which for $\kappa_{\text{feat}} \geq -1$ covers the whole stable range, the kernel sum is the left side of the secular equation, and at $\alpha_{\text{diag}} = 1$ with the pool factor of sampling without replacement it has the closed form of the Volterra kernel norm of Lee et al. [2022].

3. **Cross-mode coupling vs. one-dimensional stability (Section 4).** For Rademacher features ($\kappa_{\text{feat}} = -2$), the directional curvature $u_1^\top H_{B_t} u_1$ is deterministic, so the one-dimensional threshold along $u_1$ equals the deterministic bound $2(1+\beta)/\lambda_1$ at every batch size $B$, whereas off-diagonal mode coupling $\lambda_1\lambda_k/B$ lowers the true threshold to $O((1-\beta)B/\operatorname{tr} H)$ at high effective rank.

## 2 General Non-Commuting Minibatch Hessians

Write $\Sigma_0 = \mathbb{E}[w_t w_t^\top]$, $D_\Theta = \mathbb{E}[(w_{t+1}-w_t)(w_{t+1}-w_t)^\top]$, and $[X,Y] = XY - YX$.

**Theorem 1** (Stationary covariance equation)**.** *In any covariance-stationary regime of* (1) *with* $\mathbb{E}[w_t] = 0$ *and no commutation assumption on* $H_{B_t}$*:*

*(1)* $D_\Theta = \frac{\eta}{1+\beta}(H\Sigma_0 + \Sigma_0 H)$.

*(2)* $\Sigma_0$ *satisfies the exact matrix equation*

$$(H\Sigma_0 + \Sigma_0 H) - \frac{\eta}{1+\beta} H\Sigma_0 H + \frac{\eta\beta}{(1-\beta)(1-\beta^2)}[H,[H,\Sigma_0]] - \frac{\eta}{1-\beta}\mathcal{V}(\Sigma_0) = \frac{\eta}{1-\beta}\Sigma_\xi. \tag{2}$$

*Proof.* Let $A = (1+\beta)I - \eta H$ and $\Sigma_1 = \mathbb{E}[w_{t+1} w_t^\top]$. Conditioning on $\mathcal{F}_{t-1}$, the $\sigma$-field of $(w_0, w_{-1})$ and $(H_{B_s}, \xi_s)_{s \leq t-1}$, gives $\mathbb{E}[w_{t+1} \mid \mathcal{F}_{t-1}] = Aw_t - \beta w_{t-1}$; multiplying on the right by $w_t^\top$ yields $\Sigma_1 = A\Sigma_0 - \beta\Sigma_1^\top$ and, by transposition, $\Sigma_1^\top = \Sigma_0 A - \beta\Sigma_1$. Solving for $\Sigma_1$ gives

$$\Sigma_1 = \frac{A\Sigma_0 - \beta\Sigma_0 A}{1-\beta^2}, \qquad \Sigma_1 + \Sigma_1^\top = \frac{A\Sigma_0 + \Sigma_0 A}{1+\beta} = 2\Sigma_0 - \frac{\eta}{1+\beta}(H\Sigma_0 + \Sigma_0 H),$$

which gives (1) since $D_\Theta = 2\Sigma_0 - \Sigma_1 - \Sigma_1^\top$. For (2), expanding $\Sigma_0 = \mathbb{E}[w_{t+1} w_{t+1}^\top]$ with $w_{t+1} = Aw_t - \beta w_{t-1} - \eta(H_{B_t} - H)w_t - \eta\xi_t$ the cross terms vanish because $(H_{B_t} - H, \xi_t)$ is independent of $(w_t, w_{t-1})$ with zero mean and $\mathbb{E}[w_t] = 0$, which gives

$$\Sigma_0 = A\Sigma_0 A + \beta^2\Sigma_0 - \beta(A\Sigma_1 + \Sigma_1^\top A) + \eta^2\mathcal{V}(\Sigma_0) + \eta^2\Sigma_\xi.$$

Since $A^2\Sigma_0+\Sigma_0A^2-2\beta A\Sigma_0A=[A,[A,\Sigma_0]]+2(1-\beta)A\Sigma_0A$ and $[A,[A,\Sigma_0]]=\eta^2[H,[H,\Sigma_0]]$, substituting $\Sigma_1$ yields

$$(1-\beta^2)\Sigma_0-\frac{1-\beta}{1+\beta}A\Sigma_0A=-\frac{\beta\eta^2}{1-\beta^2}[H,[H,\Sigma_0]]+\eta^2\mathcal{V}(\Sigma_0)+\eta^2\Sigma_\xi.$$

Expanding $A\Sigma_0A=(1+\beta)^2\Sigma_0-\eta(1+\beta)(H\Sigma_0+\Sigma_0H)+\eta^2H\Sigma_0H$ and dividing by $\eta(1-\beta)$ gives (2). □

Let $s_t=(w_t,w_{t-1})\in\mathbb{R}^{2P}$ and $Q_t=\mathbb{E}[s_ts_t^\top]\in\mathbb{S}_+^{2P}$. The transition matrix of (1) is $F_t=\begin{pmatrix}(1+\beta)I-\eta H_{B_t} & -\beta I\\ I & 0\end{pmatrix}$, and the second-moment operator is $\mathcal{T}(Q)=\mathbb{E}[F_tQF_t^\top]$.

**Lemma 2** (Reduction to $\mathbb{S}^{2P}$ and instability criterion)**.** *The operator $\mathcal{T}$ preserves $\mathbb{S}^{2P}$ and $\mathbb{S}_+^{2P}$, satisfies $\|\mathcal{T}^k\|=\|\mathcal{T}^k(I)\|$ in the operator norm induced by the spectral norm on $\mathbb{R}^{2P\times2P}$, and has $\rho(\mathcal{T})=\rho(\mathcal{T}|_{\mathbb{S}^{2P}})$. When $\Sigma_\xi\succ0$ and either $\xi_t$ is independent of $H_{B_t}$ or $\mathbb{E}[s_0]=0$, $\sup_{t\geq0}\mathbb{E}\|w_t\|^2<\infty$ if and only if $\rho(\mathcal{T})<1$; if $\rho(\mathcal{T})\geq1$, then $\mathbb{E}\|w_t\|^2+\mathbb{E}\|w_{t-1}\|^2\geq ct$ for all $t\geq1$ with $c>0$.*

The equivalence of bounded second moments and $\rho(\mathcal{T})<1$ is standard [Costa et al., 2005, Chapter 3]; Lemma 2 adds the linear lower bound.

*Proof.* Since $\mathcal{T}(Q)^\top=\mathcal{T}(Q^\top)$ and $\mathcal{T}^k$ is positive in Kraus form, the Russo–Dye theorem gives $\|\mathcal{T}^k\|=\|\mathcal{T}^k(I)\|$ with $\mathcal{T}^k(I)\in\mathbb{S}_+^{2P}$, so $\rho(\mathcal{T})=\lim_{k\to\infty}\|\mathcal{T}^k(I)\|^{1/k}=\rho(\mathcal{T}|_{\mathbb{S}^{2P}})$. Under the noise assumption, $Q_{t+1}=\mathcal{T}(Q_t)+G$ with $G=\mathrm{blkdiag}(\eta^2\Sigma_\xi,0)$. If $\rho(\mathcal{T})<1$, $Q_t=\mathcal{T}^t(Q_0)+\sum_{s<t}\mathcal{T}^s(G)$ is bounded. Conversely, by the finite-dimensional Krein–Rutman theorem [Berman and Plemmons, 1994, Thm. 1.3.2], the adjoint $\mathcal{T}^*(Y)=\mathbb{E}[F_t^\top YF_t]$ on $\mathbb{S}^{2P}$ has a nonzero eigenmatrix $W\in\mathbb{S}_+^{2P}$ with eigenvalue $\rho(\mathcal{T})$. Its upper-left block $W_{11}$ is nonzero (otherwise $W=\mathrm{blkdiag}(0,W_{22})$ and $F_t^\top WF_t=\mathrm{blkdiag}(W_{22},0)$ would force $W=0$). Thus $g=\mathrm{tr}(WG)=\eta^2\,\mathrm{tr}(W_{11}\Sigma_\xi)>0$, and $\mathrm{tr}(WQ_{t+1})=\rho(\mathcal{T})\,\mathrm{tr}(WQ_t)+g\geq\mathrm{tr}(WQ_t)+g$ whenever $\rho(\mathcal{T})\geq1$, so $\mathrm{tr}(WQ_t)\geq tg$ by induction from $\mathrm{tr}(WQ_0)\geq0$, giving $\mathrm{tr}\,Q_t\geq(g/\,\mathrm{tr}\,W)t$. □

We now reduce $\rho(\mathcal{T})<1$ from $\mathbb{S}^{2P}$ to $\mathbb{S}^P$. Define the deterministic operator

$$\mathcal{L}_{\eta,\beta}(X)\triangleq(HX+XH)-\frac{\eta}{1+\beta}HXH+\frac{\eta\beta}{(1-\beta)(1-\beta^2)}[H,[H,X]]$$

on $\mathbb{S}^P$. In the eigenbasis of $H$, $\mathcal{L}_{\eta,\beta}$ is diagonal, multiplying entry $(j,k)$ by

$$\ell_{jk}(\eta,\beta)=(\lambda_j+\lambda_k)-\frac{\eta\lambda_j\lambda_k}{1+\beta}+\frac{\eta\beta(\lambda_j-\lambda_k)^2}{(1-\beta)(1-\beta^2)},$$

which is strictly positive for all $\eta\in(0,\eta_{\mathrm{det}})$, where $\eta_{\mathrm{det}}\triangleq\frac{2(1+\beta)}{\lambda_1}$.

**Theorem 3** (Exact mean-square threshold for arbitrary per-sample Hessians)**.** *Let $H\succ0$ and $\mathcal{V}(X)=\mathbb{E}[(H_{B_t}-H)X(H_{B_t}-H)]\neq0$. For $\eta\in(0,\eta_{\mathrm{det}})$, set $\mathcal{K}_{\eta,\beta}(X)\triangleq\frac{\eta}{1-\beta}\mathcal{L}_{\eta,\beta}^{-1}(\mathcal{V}(X))$ on $\mathbb{S}^P$, and define*

$$\mathcal{L}_0(X)\triangleq HX+XH,\qquad\mathcal{B}_\beta(X)\triangleq\frac{1}{1+\beta}HXH-\frac{\beta}{(1-\beta)(1-\beta^2)}[H,[H,X]]+\frac{1}{1-\beta}\mathcal{V}(X).$$

*(1) For every $\eta\in(0,\eta_{\mathrm{det}})$, $\mathcal{L}_{\eta,\beta}^{-1}$ and $\mathcal{K}_{\eta,\beta}$ preserve the cone $\mathbb{S}_+^P$, $\rho(\mathcal{K}_{\eta,\beta})$ is the top eigenvalue of the variational quotient*

$$\rho(\mathcal{K}_{\eta,\beta})=\frac{1}{1-\beta}\max_{X\in\mathbb{S}^P\setminus\{0\}}\frac{\mathrm{tr}\left(X\mathcal{V}(X)\right)}{\sum_{j,k=1}^P\left(\ell_{jk}(\eta,\beta)/\eta\right)(u_j^\top Xu_k)^2},\tag{3}$$

*which increases strictly and continuously with $\eta\in(0,\eta_{\mathrm{det}})$, and $\rho(\mathcal{T})<1\iff\rho(\mathcal{K}_{\eta,\beta})<1$.*

*(2) The mean-square stability threshold* $\eta^\star(\beta) \triangleq \sup\{\eta > 0 : \rho(\mathcal{T}) < 1\}$ *is given exactly by*

$$\eta^\star(\beta) = \frac{1}{\lambda_{\max}\big(\mathcal{L}_0^{-1/2}\mathcal{B}_\beta\mathcal{L}_0^{-1/2}\big)} \le \eta_{\det}, \tag{4}$$

*where* $\mathcal{L}_0^{-1/2}\mathcal{B}_\beta\mathcal{L}_0^{-1/2}$ *is self-adjoint on* $(\mathbb{S}^P, \langle X, Y\rangle_F = \operatorname{tr}(XY))$. *If* $(H_{B_t} - H)P_1 \not\equiv 0$ *for the orthogonal projector* $P_1$ *onto* $\ker(H - \lambda_1 I)$, *then* $\lim_{\eta\uparrow\eta_{\det}} \rho(\mathcal{K}_{\eta,\beta}) = +\infty$ *and* $\eta^\star(\beta) < \eta_{\det}$ *is the unique root of* $\rho(\mathcal{K}_{\eta,\beta}) = 1$.

As $(1-\beta)\operatorname{tr}(X\mathcal{B}_\beta(X))/\operatorname{tr}(X\mathcal{L}_0(X))$ is the quotient $q_{H,\beta}(X)$ of Cai et al. [2026], (4) is their Theorem F.1.

*Proof.* (1) Decompose $\mathcal{T} = \mathcal{T}_0 + \Delta$ on $\mathbb{S}^{2P}$ into the mean part $\mathcal{T}_0(Q) = \bar{F}Q\bar{F}^\top$ ($\bar{F} = \mathbb{E}[F_t]$) and the noise part $\Delta(Q) = \operatorname{blkdiag}(\eta^2\mathcal{V}(Q_{11}), 0)$. For $\eta \in (0, \eta_{\det})$, $\rho(\mathcal{T}_0) = \rho(\bar{F})^2 < 1$, and $(I - \mathcal{T}_0)^{-1} = \sum_{t\ge 0}\mathcal{T}_0^t$ preserves $\mathbb{S}_+^{2P}$. For any $Y \in \mathbb{S}_+^P$, the matrix $Q = (I - \mathcal{T}_0)^{-1}\operatorname{blkdiag}(\eta^2 Y, 0) \in \mathbb{S}_+^{2P}$ is the stationary covariance of (1) with $\mathcal{V} = 0$ and $\Sigma_\xi = Y$, so Theorem 1(2) gives $Q_{11} = \frac{\eta}{1-\beta}\mathcal{L}_{\eta,\beta}^{-1}(Y) \in \mathbb{S}_+^P$. Thus $\mathcal{L}_{\eta,\beta}^{-1}$ preserves $\mathbb{S}_+^P$. Writing $\Pi(Q) = Q_{11}$ and $\mathcal{J}(X) = (I - \mathcal{T}_0)^{-1}\operatorname{blkdiag}(\eta^2\mathcal{V}(X), 0)$, we obtain $(I - \mathcal{T}_0)^{-1}\Delta = \mathcal{J} \circ \Pi$ and $\Pi \circ \mathcal{J} = \mathcal{K}_{\eta,\beta}$. Since $\mathcal{J} \circ \Pi$ and $\Pi \circ \mathcal{J}$ share the same nonzero eigenvalues, $\rho((I - \mathcal{T}_0)^{-1}\Delta) = \rho(\mathcal{K}_{\eta,\beta})$, and by regular splitting on the cone $\mathbb{S}_+^{2P}$ [Berman and Plemmons, 1994], $\rho(\mathcal{T}) < 1 \iff \rho((I - \mathcal{T}_0)^{-1}\Delta) < 1 \iff \rho(\mathcal{K}_{\eta,\beta}) < 1$. Because $\mathcal{N}_{\eta,\beta} = \eta\mathcal{L}_{\eta,\beta}^{-1} \succ 0$ and $\mathcal{V}$ are self-adjoint on $(\mathbb{S}^P, \langle\cdot,\cdot\rangle_F)$, $\mathcal{K}_{\eta,\beta}$ is similar to the symmetric operator $\frac{1}{1-\beta}\mathcal{N}_{\eta,\beta}^{1/2}\mathcal{V}\mathcal{N}_{\eta,\beta}^{1/2}$ and preserves $\mathbb{S}_+^P$; by the Krein–Rutman theorem [Krein and Rutman, 1948], $\rho(\mathcal{K}_{\eta,\beta}) = \lambda_{\max}(\mathcal{K}_{\eta,\beta})$ is given by (3). Since $\operatorname{tr}(I_P\mathcal{V}(I_P)) = \mathbb{E}\|H_{B_t} - H\|_F^2 > 0$ and each weight $\ell_{jk}(\eta,\beta)/\eta = (\lambda_j + \lambda_k)/\eta - \frac{\lambda_j\lambda_k}{1+\beta} + \frac{\beta(\lambda_j - \lambda_k)^2}{(1-\beta)(1-\beta^2)}$ decreases strictly with $\eta$, $\rho(\mathcal{K}_{\eta,\beta})$ is strictly increasing in $\eta$.

(2) Since $\mathcal{L}_{\eta,\beta} = \mathcal{L}_0 - \eta\mathcal{B}_\beta + \frac{\eta}{1-\beta}\mathcal{V}$, we have $\rho(\mathcal{K}_{\eta,\beta}) < 1 \iff \lambda_{\max}(\mathcal{N}_{\eta,\beta}^{1/2}\frac{\mathcal{V}}{1-\beta}\mathcal{N}_{\eta,\beta}^{1/2}) < 1 \iff \mathcal{L}_0 - \eta\mathcal{B}_\beta \succ 0 \iff \eta\lambda_{\max}(\mathcal{L}_0^{-1/2}\mathcal{B}_\beta\mathcal{L}_0^{-1/2}) < 1$. For $\eta \ge \eta_{\det}$, $\mathcal{T} \succeq \mathcal{T}_0$ on $\mathbb{S}_+^{2P}$ gives $\rho(\mathcal{T}) \ge \rho(\mathcal{T}_0) \ge 1$; testing $X = u_1u_1^\top$ gives $\operatorname{tr}(X\mathcal{B}_\beta(X))/\operatorname{tr}(X\mathcal{L}_0(X)) \ge 1/\eta_{\det}$, so (4) holds in all cases. As $\eta \uparrow \eta_{\det}$, $\ell_{11}(\eta,\beta)/\eta \asymp \eta_{\det} - \eta \downarrow 0$, so testing $X = sP_1 + \mathcal{V}(P_1)$ with $s = (\ell_{11}/\eta)^{-1/3}$ in (3) gives $\rho(\mathcal{K}_{\eta,\beta}) \to +\infty$ whenever $(H_{B_t} - H)P_1 \not\equiv 0$ (then $\operatorname{tr}\mathcal{V}(P_1) > 0$, the numerator grows at least like $2s\|\mathcal{V}(P_1)\|_F^2$, and the denominator stays bounded). □

At $\beta = 0$, (4) has the form of Theorem 5 of Mulayoff and Michaeli [2024], which they prove at interpolating minima for sampling without replacement (with a pseudo-inverse); here $H \succ 0$. For minibatches of size $B$ drawn from $N$ fixed per-sample Hessians $H_1, \dots, H_N$ with mean $H$, $\mathcal{V}(X) = p\,\frac{1}{N}\sum_{i=1}^N (H_i - H)X(H_i - H)$ with $p = 1/B$ for sampling with replacement and $p = \frac{N-B}{B(N-1)}$ for sampling without replacement.

**Corollary 4** (Strict monotonicity of the effective step at the threshold)**.** *If* $H \succ 0$ *and* $\mathcal{V}$ *does not depend on* $\beta$, *the effective threshold* $\eta_{\text{eff}}^\star(\beta) \triangleq \eta^\star(\beta)/(1-\beta)$ *increases strictly with* $\beta \in [0, 1)$ *and satisfies*

$$\frac{1}{\eta^\star(0)} - \frac{1-\beta}{\eta^\star(\beta)} \ge \frac{\beta}{1+\beta}\lambda_{\min}(H). \tag{5}$$

*Proof.* By (4), $1/\eta_{\text{eff}}^\star(\beta) = \lambda_{\max}(\mathcal{L}_0^{-1/2}\mathcal{A}_\beta\mathcal{L}_0^{-1/2})$ with $\mathcal{A}_\beta = (1-\beta)\mathcal{B}_\beta$. The noise operator $\mathcal{V}$ cancels in $\mathcal{B}_0 - \mathcal{A}_\beta = \frac{2\beta}{1+\beta}H(\cdot)H + \frac{\beta}{1-\beta^2}[H,[H,\cdot]]$, which is diagonal in the eigenbasis of $H$ with entries $m_{jk}(\beta) = \frac{2\beta}{1+\beta}\lambda_j\lambda_k + \frac{\beta}{1-\beta^2}(\lambda_j - \lambda_k)^2$. Since $m_{jk}(\beta)$ increases strictly on $[0,1)$ and $\lambda_j\lambda_k > 0$, $\mathcal{L}_0^{-1/2}(\mathcal{A}_{\beta_1} - \mathcal{A}_{\beta_2})\mathcal{L}_0^{-1/2} \succ 0$ for $\beta_1 < \beta_2$, so Weyl's inequality gives $\eta_{\text{eff}}^\star(\beta_2) > \eta_{\text{eff}}^\star(\beta_1)$. Using $m_{jk}(\beta) \ge \frac{\beta}{1+\beta}(\lambda_j^2 + \lambda_k^2) \ge \frac{\beta}{1+\beta}\lambda_{\min}(H)(\lambda_j + \lambda_k)$ gives $\mathcal{L}_0^{-1/2}(\mathcal{B}_0 - \mathcal{A}_\beta)\mathcal{L}_0^{-1/2} \succeq \frac{\beta}{1+\beta}\lambda_{\min}(H)I$ and hence (5). □

## 3 Random-Feature Minibatches and the Expected Loss

Suppose now that $H_{B_t} = \frac{1}{B}\sum_{b=1}^{B} z_{t,b}z_{t,b}^\top$ with $z_{t,b} = H^{1/2}\omega_{t,b}$ drawn afresh and independently for every $(t,b)$, where the coordinates $\omega_k$ of $\omega$ in the eigenbasis of $H$ are independent with $\mathbb{E}[\omega_k] = 0$, $\mathbb{E}[\omega_k^2] = 1$, and $\mathbb{E}[\omega_k^4] = 3 + \kappa_{\text{feat}}$ for $\kappa_{\text{feat}} \geq -2$ ($\kappa_{\text{feat}} = 0$ for Gaussian features, $\kappa_{\text{feat}} = -2$ for Rademacher features $\omega_k \in \{\pm 1\}$). Write $\kappa_\beta = \frac{1+\beta}{1-\beta}$ and $\alpha_{\text{diag}} = 1 + p\kappa_\beta(1+\kappa_{\text{feat}})$, with $p = 1/B$.

**Theorem 5** (Rank-one secular equation)**.** *In the random-feature model above with $\sigma_{\xi,k}^2 = u_k^\top \Sigma_\xi u_k > 0$, the action of $\mathcal{V}$ on $X \in \mathbb{S}^P$ in the eigenbasis of $H$ is*

$$\mathcal{V}(X)_{kk} = p\lambda_k \operatorname{tr}(HX) + p(1+\kappa_{\text{feat}})\lambda_k^2 X_{kk}, \qquad \mathcal{V}(X)_{jk} = p\lambda_j\lambda_k X_{jk} \quad (j \neq k). \tag{6}$$

*Consequently, the off-diagonal entries of $\Sigma_0$ decouple from the diagonal, the normalized diagonal step variances $c_k = \eta^{-2}u_k^\top D_\Theta u_k$ solve, for $\eta < \eta^\star(\beta)$, the M-matrix system*

$$\Big[I - \frac{\eta}{2(1+\beta)}\big(\alpha_{\text{diag}}\Lambda + p\kappa_\beta\lambda\mathbf{1}^\top\big)\Big]c = \frac{(\sigma_{\xi,k}^2)_{k=1}^P}{1-\beta^2}, \qquad \Lambda = \operatorname{diag}(\lambda_1,\dots,\lambda_P),$$

*and the mean-square threshold $\eta^\star(\beta)$ is the unique root with $\eta\lambda_1\max(\alpha_{\text{diag}},0) < 2(1+\beta)$ of the secular equation*

$$\frac{\eta p}{2(1-\beta)}\sum_{k=1}^{P}\frac{\lambda_k}{1 - \frac{\eta\lambda_k}{2(1+\beta)}\alpha_{\text{diag}}} = 1. \tag{7}$$

*Moreover, $\eta^\star(\beta) \leq \frac{2(1+\beta)}{\lambda_1\max(\alpha_{\text{diag}},1)}$, strictly when $\alpha_{\text{diag}} > 1$, and $\mu^\star \triangleq 2(1+\beta)/\eta^\star(\beta)$ satisfies the two-sided bounds*

$$\max(\alpha_{\text{diag}},0)\lambda_1 + p\kappa_\beta\operatorname{tr}H \geq \mu^\star \geq \max\Big(\alpha_{\text{diag}}\lambda_1 + p\kappa_\beta\lambda_1,\ \min(\alpha_{\text{diag}},0)\lambda_1 + p\kappa_\beta\operatorname{tr}H\Big). \tag{8}$$

*Proof.* Since $\operatorname{Cov}(\omega_j\omega_l, \omega_m\omega_k)$ is 1 when $j = m \neq l = k$ or $j = k \neq l = m$, $2+\kappa_{\text{feat}}$ when $j = l = m = k$, and 0 otherwise, expanding $\mathcal{V}(X) = p\,\mathbb{E}[(zz^\top - H)X(zz^\top - H)]$ gives (6). By (6), $\mathcal{L}_0$ and $\mathcal{B}_\beta$ leave the diagonal subspace of $\mathbb{S}^P$ invariant and multiply each off-diagonal entry $(j,k)$ $(j \neq k)$ by $\lambda_j + \lambda_k$ and by $b_{jk} = \frac{1+p\kappa_\beta}{1+\beta}\lambda_j\lambda_k - \frac{\beta(\lambda_j-\lambda_k)^2}{(1-\beta)(1-\beta^2)} \leq \frac{1+p\kappa_\beta}{1+\beta}\lambda_j\lambda_k$. On the diagonal subspace $X = \operatorname{diag}(\gamma_0)$ with $\gamma_{0,k} = \frac{\eta(1+\beta)}{2\lambda_k}c_k$, the $(k,k)$ entry of (2) reads $\big[I - \frac{\eta}{2(1+\beta)}\mathcal{M}\big]c = \frac{(\sigma_{\xi,k}^2)_{k=1}^P}{1-\beta^2}$, where $\mathcal{M} = \alpha_{\text{diag}}\Lambda + p\kappa_\beta\lambda\mathbf{1}^\top$ is similar to $\widetilde{\mathcal{M}} = \alpha_{\text{diag}}\Lambda + p\kappa_\beta\sqrt{\lambda}\sqrt{\lambda}^\top$. The off-diagonal entries are never less stable than the diagonal block: for $j \neq k$, the test vector $v = (e_j + e_k)/\sqrt{2}$ gives $\lambda_{\max}(\widetilde{\mathcal{M}}) \geq v^\top\widetilde{\mathcal{M}}v = (\alpha_{\text{diag}} + p\kappa_\beta)\frac{\lambda_j+\lambda_k}{2} + p\kappa_\beta\sqrt{\lambda_j\lambda_k} \geq (1+p\kappa_\beta)\frac{2\lambda_j\lambda_k}{\lambda_j+\lambda_k} \geq \frac{2(1+\beta)\,b_{jk}}{\lambda_j+\lambda_k}$, because $\alpha_{\text{diag}} + p\kappa_\beta = 1 + p\kappa_\beta(2+\kappa_{\text{feat}}) \geq 1$ (this is where $\kappa_{\text{feat}} \geq -2$ is used) and the arithmetic and geometric means of $\lambda_j, \lambda_k$ are at least their harmonic mean. Hence the top eigenvalue of $\mathcal{L}_0^{-1/2}\mathcal{B}_\beta\mathcal{L}_0^{-1/2}$ is that of the diagonal block, $\lambda_{\max}(\widetilde{\mathcal{M}})/(2(1+\beta))$, and Theorem 3 gives $\eta^\star(\beta) = 2(1+\beta)/\lambda_{\max}(\widetilde{\mathcal{M}})$. By the rank-one determinant identity

$$\det(\mu I - \widetilde{\mathcal{M}}) = \prod_{k=1}^{P}(\mu - \alpha_{\text{diag}}\lambda_k)\left(1 - p\kappa_\beta\sum_{k=1}^{P}\frac{\lambda_k}{\mu - \alpha_{\text{diag}}\lambda_k}\right),$$

$\mu^\star = \lambda_{\max}(\widetilde{\mathcal{M}})$ is the unique root of (7) with $\mu = 2(1+\beta)/\eta > \max(\alpha_{\text{diag}},0)\lambda_1$; since $e_1^\top\widetilde{\mathcal{M}}e_1 = (\alpha_{\text{diag}} + p\kappa_\beta)\lambda_1 \geq \lambda_1$, also $\mu^\star \geq \lambda_1$, which gives the upper bound on $\eta^\star(\beta)$. Replacing $\alpha_{\text{diag}}\lambda_k$ in each denominator of (7) by $\max(\alpha_{\text{diag}},0)\lambda_1$ or $\min(\alpha_{\text{diag}},0)\lambda_1$, or keeping only the $k=1$ term, gives (8). □

**Proposition 6** (Finite-$P$ convolution equation for the expected loss)**.** *In the setting of Theorem 5 with $\Sigma_\xi$ diagonal in the eigenbasis of $H$, $w_0 = w_{-1} = 0$, and $\eta < \eta_{\text{det}}$, let $g_k(0) = 1$, $g_k(1) = 1+\beta-\eta\lambda_k$, and $g_k(\tau+$*

$1) = (1+\beta-\eta\lambda_k)g_k(\tau)-\beta g_k(\tau-1)$ *be the deterministic impulse response of mode* $k$*, let* $\psi_k(\tau) = \eta^2 g_k(\tau)^2$*, and let* $h_k$ *solve the scalar convolution equation* $h_k(\tau) = \psi_k(\tau) + p(1+\kappa_{\text{feat}})\lambda_k^2 \sum_{u<\tau} \psi_k(\tau-1-u)h_k(u)$*. Then twice the expected loss,* $\ell_t \triangleq \operatorname{tr}(H\mathbb{E}[w_t w_t^\top])$*, satisfies the exact discrete convolution equation*

$$\ell_t = f_t + \sum_{s=0}^{t-1} K_{t-1-s}\,\ell_s, \qquad K_\tau = p\sum_{k=1}^{P} \lambda_k^2 h_k(\tau), \qquad f_t = \sum_{k=1}^{P} \lambda_k \sigma_{\xi,k}^2 \sum_{\tau<t} h_k(\tau). \tag{9}$$

*Whenever* $\eta\lambda_1 \max(\alpha_{\text{diag}}, 2-\alpha_{\text{diag}}) < 2(1+\beta)$*, the kernel* $K_\tau$ *is summable and its sum is the left-hand side of the secular equation (7):*

$$\sum_{\tau=0}^{\infty} K_\tau = \frac{\eta p}{2(1-\beta)} \sum_{k=1}^{P} \frac{\lambda_k}{1 - \frac{\eta\lambda_k}{2(1+\beta)}\alpha_{\text{diag}}}. \tag{10}$$

*For* $\kappa_{\text{feat}} \geq -1$ *this condition holds on* $(0, \eta^\star(\beta)]$*, so* $\eta^\star(\beta)$ *is the root of* $\sum_\tau K_\tau = 1$*. For* $\kappa_{\text{feat}} \in [-2, -1)$ *it can fail below* $\eta^\star(\beta)$ *(e.g.* $P=2$*,* $\lambda=(1, 0.05)$*,* $\beta=0$*,* $B=1$*, Rademacher:* $\eta^\star = 2/1.05$*, condition only for* $\eta<1$*); the threshold is still that of Theorem 5.*

*Proof.* Iterating $Q_{t+1} = \mathcal{T}_0(Q_t) + \Delta(Q_t) + G$ from $Q_0 = 0$ gives $Q_t = \sum_{s\leq t} \mathcal{T}_0^{t-1-s}(\Delta(Q_s) + G)$. Since $(\bar{F}^\tau)_{11} = \operatorname{diag}(g_k(\tau))$ in the eigenbasis of $H$, (6) implies that $x_{t,k} = (\mathbb{E}[w_t w_t^\top])_{kk}$ obeys

$$x_{t,k} = \sum_{s=0}^{t-1} \psi_k(t-1-s)\big(\sigma_{\xi,k}^2 + p\lambda_k \ell_s + p(1+\kappa_{\text{feat}})\lambda_k^2 x_{s,k}\big).$$

By definition of $h_k$, $x_{t,k} = \sum_{s<t} h_k(t-1-s)(\sigma_{\xi,k}^2 + p\lambda_k\ell_s)$; multiplying by $\lambda_k$ and summing over $k$ yields (9). By the Yule–Walker identity for the second-order autoregression $g_k$, $\Psi_k \triangleq \sum_{\tau=0}^{\infty} \psi_k(\tau) = \frac{\eta}{2(1-\beta)\lambda_k(1-\eta\lambda_k/(2(1+\beta)))}$. When $|p(1+\kappa_{\text{feat}})\lambda_k^2|\Psi_k < 1$ (which is $\eta\lambda_k \max(\alpha_{\text{diag}}, 2-\alpha_{\text{diag}}) < 2(1+\beta)$), the Neumann series for $h_k$ converges absolutely with $\sum_{\tau=0}^{\infty} h_k(\tau) = \Psi_k/\big(1-p(1+\kappa_{\text{feat}})\lambda_k^2\Psi_k\big)$, and summing $p\lambda_k^2\sum_\tau h_k(\tau)$ over $k$ gives (10). For $\kappa_{\text{feat}} \geq -1$, $\alpha_{\text{diag}} \geq 1$ and $\mu^\star > \alpha_{\text{diag}}\lambda_1$ give $\eta^\star\lambda_1\alpha_{\text{diag}} < 2(1+\beta)$. In the example, $\alpha_{\text{diag}} = 0$, $\mu^\star = 1.05$, and $|p(1+\kappa_{\text{feat}})\lambda_1^2|\Psi_1 < 1$ only for $\eta < 1$. □

**Connection to the high-dimensional Volterra equation of Lee et al. [2022] and momentum LMS.** For large $n$ and $d$ under orthogonal invariance of the data, Lee et al. [2022] describe the loss of minibatch SGD with momentum by a Volterra equation; their Proposition 2 gives conditions, among them $\gamma < (1+\Delta)/(\zeta\sigma_{\max}^2)$, for $\|\mathcal{K}\| < 1$. Under the dictionary $\eta = \gamma B$, $\beta = \Delta$, $\lambda_j = \sigma_j^2/n$ and $\zeta = B/n$ (their update sums the $B$ gradients whereas (1) averages them), the kernel sum (10) at $\alpha_{\text{diag}} = 1$, with $p$ replaced by $(1-\zeta)/B$, the large-$n$ form of the without-replacement factor $\frac{n-B}{B(n-1)}$, becomes

$$\frac{1}{n}\sum_{j=1}^{P} \frac{(1-\zeta)\gamma\sigma_j^2(1+\Delta)}{(1-\Delta)(2(1+\Delta) - \gamma\zeta\sigma_j^2)},$$

which is the closed-form kernel norm $\|\mathcal{K}\|$ in the proof of their Proposition 2; their step condition reads $\eta\lambda_1 < 1+\beta$, half the deterministic threshold. The two formulas coincide, not the two settings: the pool factor is outside Theorem 5 (fresh features), both data classes contain Gaussian data, and their statement is asymptotic while Proposition 6 is exact at finite $P$. With $B = \zeta n$ at fixed $\beta$ and $\zeta$, $\alpha_{\text{diag}} - 1 = p\kappa_\beta(1+\kappa_{\text{feat}}) = O(1/n)$; at finite $P$ and $B$, $\alpha_{\text{diag}}$ is what Theorem 5 adds to that formula. At the opposite extreme $B=1$ with Gaussian inputs ($\kappa_{\text{feat}} = 0$, $\alpha_{\text{diag}} = 1+\kappa_\beta$), (7) with $\eta = 2\mu$ and $\beta = \alpha$ in their notation coincides with the Schur–Cohn condition $Q(1) > 0$ derived as necessary for momentum LMS by Roy and Shynk [1990, eqs. (31)–(33)]; Theorem 5 proves that it is also sufficient for mean-square stability when $2\mu\lambda_k < 1-\alpha^2$ for every $k$, the region below the first pole of (7).

## 4 Asymptotic Regimes and Rademacher Separation

The sandwich (8) captures two distinct regimes of the secular equation (7):

- **Large batch / low effective rank** ($p\kappa_\beta \operatorname{tr} H/\lambda_1 \ll 1$)**:** $\eta^\star(\beta) = \frac{2(1+\beta)}{\lambda_1}\big(1 - p\kappa_\beta(2+\kappa_{\text{feat}}) + O(p^2)\big)$ for fixed $H$ with $\lambda_1 > \lambda_2$ as $p \to 0$, the $O(p^2)$ term depending on the spectral gap through $\sum_{k\geq 2} \lambda_k/(\lambda_1 - \lambda_k)$, recovering the deterministic ceiling $2(1+\beta)/\lambda_1$.
- **Small batch / high effective rank** ($p\kappa_\beta \operatorname{tr} H/\lambda_1 \gg 1$, $\lambda_1/\operatorname{tr} H \to 0$)**:** $\eta^\star(\beta) = \frac{2(1-\beta)}{p \operatorname{tr} H}\big(1 + o(1)\big)$, governed by $\operatorname{tr} H$ rather than $\lambda_1$; Wang et al. [2026] obtain a critical-rate scaling of this form, $\min\{1, B(1-\beta)\}$, for one-pass power-law kernel regression.

Theorem 5 has a consequence for Rademacher features ($\kappa_{\text{feat}} = -2$). Along the top eigenvector $u_1$, the directional curvature is $u_1^\top H_{B_t} u_1 = \lambda_1 \frac{1}{B}\sum_{b=1}^{B} \omega_{t,b,1}^2 \equiv \lambda_1$ because $\omega_1^2 \equiv 1$. Hence the directional curvature variance $\nu_{11,B}^2 = \operatorname{Var}(u_1^\top H_{B_t} u_1)$ vanishes identically, and the one-dimensional stability threshold along $\operatorname{span}(u_1)$ [Wu et al., 2018, Andreyev et al., 2026],

$$\eta_{\text{crit}}^{\text{1D}}(\beta, B) = \frac{2(1+\beta)/\lambda_1}{1 + \kappa_\beta \nu_{11,B}^2/\lambda_1^2}, \tag{11}$$

equals the deterministic threshold $2(1+\beta)/\lambda_1$ for every batch size $B$. Yet off-diagonal curvature fluctuations $(u_1^\top H_{B_t} u_k)^2$ have mean $\lambda_1\lambda_k/B > 0$ for $k \geq 2$ and couple the modes: by (8), the true threshold satisfies $\eta^\star(\beta) \leq \frac{2(1+\beta)}{\lambda_1 + \kappa_\beta \operatorname{tr}(H_\perp)/B}$ for $B \leq \kappa_\beta$, where $\operatorname{tr}(H_\perp) = \operatorname{tr} H - \lambda_1$.

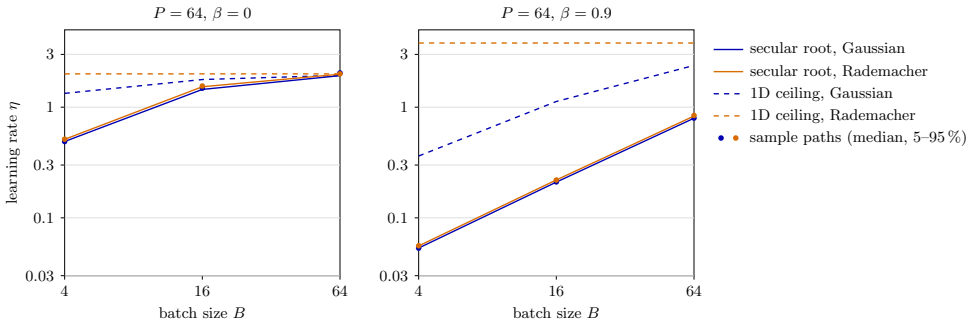



Figure 1: Mean-square stability threshold $\eta^\star$ against batch size $B$ for $H = \operatorname{diag}(k^{-1/2})_{k=1}^{P}$ at $P = 64$, $\beta = 0$ (left) and $\beta = 0.9$ (right). Solid curves: secular root of Theorem 5; dashed curves: one-dimensional threshold (11) along $u_1$; dots: median empirical stability threshold over 8 sample-path bisection seeds (5–95% range); the Gaussian dot at $\beta = 0$, $B = 64$ is at the bisection cap (Section 4). Blue: Gaussian features ($\kappa_{\text{feat}} = 0$); orange: Rademacher features ($\kappa_{\text{feat}} = -2$), whose one-dimensional threshold along $u_1$ is flat at $2(1+\beta)/\lambda_1$.

Figure 1 and Table 1 illustrate Theorem 5 on $H = \operatorname{diag}(k^{-1/2})_{k=1}^{P}$. At $P = 256$, $\beta = 0.9$, $B = 4$ with Rademacher features, the one-dimensional threshold along $u_1$ is $\eta_{\text{crit}}^{\text{1D}} = 3.8$, whereas the secular root is $\eta^\star = 0.0263$ and the empirical sample-path ceiling is 0.0264; the one-dimensional threshold exceeds the secular root by a factor 144.5. Sample paths need not track the mean square: at $P = 64$, $\beta = 0$, $B = 64$ with Gaussian features the median bisection run stops at its cap $1.02 \cdot 2/\lambda_1$, above the secular root 1.9268 (median 2.003 with a raised cap), consistent with stable typical paths and a mean square diverging through rare paths; an importance-resampling estimate of the growth of $\mathbb{E}\|w_t\|^2$ crosses 1 at a median step (8 seeds) 0.32% above the secular root. Code and numerical verification scripts for all results in this note are available at `https://github.com/apis-os/hb-sgd-mean-square`.

Table 1: Comparison of the one-dimensional threshold (11), the sandwich bounds (8), the exact secular root (7), and the median empirical sample-path threshold on $H = \mathrm{diag}(k^{-1/2})_{k=1}^{P}$ at $\beta = 0.9$. Across all 36 parameter cells tested ($P \in \{64, 256, 1024\}$, $\beta \in \{0, 0.9\}$, $B \in \{4, 16, 64\}$), the crossing $\rho(\mathcal{T}) = 1$ of the exact second-moment operator agrees with the secular root to at most $6.11 \times 10^{-10}$ relative error, with the operator built from (6). The last column divides the one-dimensional threshold by the unrounded secular root.

| $P$ | $B$ | Features | $\eta_{\mathrm{crit}}^{\mathrm{1D}}$ | Lower (8) | Secular $\eta^\star$ | Upper (8) | Empirical | $\eta_{\mathrm{crit}}^{\mathrm{1D}}/\eta^\star$ |
|---|---|---|---|---|---|---|---|---|
| 64 | 4 | Gaussian | 0.3619 | 0.0506 | 0.0533 | 0.0548 | 0.0532 | 6.8× |
| 64 | 4 | Rademacher | 3.8000 | 0.0548 | 0.0558 | 0.0579 | 0.0559 | 68.2× |
| 64 | 16 | Gaussian | 1.1259 | 0.1946 | 0.2103 | 0.2191 | 0.2113 | 5.4× |
| 64 | 16 | Rademacher | 3.8000 | 0.2191 | 0.2199 | 0.2215 | 0.2196 | 17.3× |
| 64 | 64 | Gaussian | 2.3843 | 0.6747 | 0.7948 | 0.8766 | 0.7971 | 3.0× |
| 64 | 64 | Rademacher | 3.8000 | 0.7543 | 0.8315 | 0.8766 | 0.8428 | 4.6× |
| 256 | 4 | Gaussian | 0.3619 | 0.0252 | 0.0260 | 0.0262 | 0.0261 | 13.9× |
| 256 | 4 | Rademacher | 3.8000 | 0.0262 | 0.0263 | 0.0269 | 0.0264 | 144.5× |
| 256 | 16 | Gaussian | 1.1259 | 0.0987 | 0.1034 | 0.1047 | 0.1039 | 10.9× |
| 256 | 16 | Rademacher | 3.8000 | 0.1047 | 0.1048 | 0.1052 | 0.1051 | 36.3× |

# References


Arseniy Andreyev, Advikar Ananthkumar, Marc Walden, Tomaso Poggio, and Pierfrancesco Beneventano. Momentum further constrains sharpness at the edge of stochastic stability. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, volume 306 of *Proceedings of Machine Learning Research*, pages 2831–2871, 2026.

Abraham Berman and Robert J. Plemmons. *Nonnegative Matrices in the Mathematical Sciences*. Classics in Applied Mathematics. SIAM, 1994.

Yuhang Cai, Haodong Wen, Haofeng Huang, Deyi Liu, Yiyuan Ma, and Kaifeng Lyu. Does LLM pre-training typically occur at the edge of stability? *arXiv preprint arXiv:2610.36778*, 2026.

Andrew Nicholas Cheng, Kiwon Lee, and Courtney Paquette. High dimensional first order mini-batch algorithms on quadratic problems. In *OPT 2024: Optimization for Machine Learning (NeurIPS 2024 Workshop)*, 2024. URL https://openreview.net/forum?id=jDttnKBKUL.

Oswaldo L. V. Costa, Marcelo D. Fragoso, and Ricardo P. Marques. *Discrete-Time Markov Jump Linear Systems*. Probability and Its Applications. Springer, London, 2005. doi: 10.1007/b138575.

Mark G. Krein and Moisei A. Rutman. Linear operators leaving invariant a cone in a Banach space. *Uspekhi Matematicheskikh Nauk*, 3(1):3–95, 1948. English translation: American Mathematical Society Translations, Series 1, Volume 10, pages 199–325, 1962.

Kiwon Lee, Andrew N. Cheng, Elliot Paquette, and Courtney Paquette. Trajectory of mini-batch momentum: Batch size saturation and convergence in high dimensions. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2206.01029.

Rotem Mulayoff and Tomer Michaeli. Exact mean square linear stability analysis for SGD. In *Proceedings of the Thirty Seventh Conference on Learning Theory (COLT)*, volume 247 of *Proceedings of Machine Learning Research*, pages 3915–3969, 2024.

Courtney Paquette and Elliot Paquette. Dynamics of stochastic momentum methods on large-scale, quadratic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. arXiv:2106.03696.

Sumit Roy and John J. Shynk. Analysis of the momentum LMS algorithm. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 38(12):2088–2098, 1990.

Nishant Sharma, Laura Gomezjurado Gonzalez, Ganesh Talluri, Sarath Chandar, and Hiroki Naganuma. Understanding stability of outer Nesterov momentum on stochastic pseudo-gradients. *arXiv preprint arXiv:2610.22623*, 2026.

Maksim Velikanov, Denis Kuznedelev, and Dmitry Yarotsky. A view of mini-batch SGD via generating functions: Conditions of convergence, phase transitions, benefit from negative momenta. In *International Conference on Learning Representations (ICLR)*, 2023.

Jia-Nan Wang, Zixun Huang, Kairui Li, Zilin Wang, and Lei Wu. Momentum in large-batch training: Polyak enlarges the critical batch size, Nesterov improves data efficiency. *arXiv preprint arXiv:2609.02728*, 2026.

Lei Wu, Chao Ma, and Weinan E. How SGD selects the global minima in over-parameterized learning: A dynamical stability perspective. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.

Kun Yuan, Bicheng Ying, and Ali H. Sayed. On the influence of momentum acceleration on online learning. *Journal of Machine Learning Research*, 17(192):1–66, 2016.